

3	(a) heat/thermal energy gained by system <i>or</i> energy transferred to system by heating plus work done on the system <i>or</i> minus work done by the system	B1 B1	[2]
	(b) (i) <i>either</i> volume decreases so work done on the system <i>or</i> small volume change so work done on system negligible (thermal) energy absorbed to break lattice structure internal energy increases	M1 M1 A1	[3]
	(ii) gas expands so work done by gas (against atmosphere) no time for thermal energy to enter or leave the gas internal energy decreases	M1 M1 A1	[3]